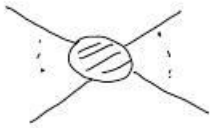
Scattering Amplitudes and Positive Geometry

Monday, May 4, 2026 12:36 PM

Amplitudes

let us consider scattering amplitudes of $n>2$ massless vector particles $A(p_i, \varepsilon_i)$.

p_i are the external momentum
ε_i polarization vectors



Any such scattering amplitude is subject to the following:

1) $\forall i,\ p_i \neq 0$
2) $p_i \cdot p_i = \varepsilon_i \cdot p_i = \varepsilon_i \cdot \varepsilon_i = 0 \quad \forall i$
3) $\sum_{i=1}^{n} p_i = 0$ (momentum conservation)

Amplitudes in general must satisfy

- $A(p_i, \kappa_i \varepsilon_i) = A(p_i, \varepsilon_i) \prod_{i=1}^{n} \kappa_i \qquad \kappa_i \in \mathbb{C}$ (linear in polarization vectors
- $A(p_i, \varepsilon_i + z_i p_i) = A(p_i, \varepsilon_i)$
- Lorentz invariant

$$A(Rp_i, R\varepsilon_i) = A(p_i, \varepsilon_i) \qquad \forall R \in So(1,3)$$

- unitarity & factorization properties $\Rightarrow$ BCFW recursion relations (2005)
  by doing complex momentum shift

$$\hat{p}_i^{\mu} = p_i^{\mu} + z r_i^{\mu} \qquad \text{s.t} \quad \sum r_i^{\mu} = 0$$
$$r_i \cdot r_j = 0$$
$$p_i \cdot r_i = 0$$

one can find

(tree level) $$A_n \equiv \hat{A}(z=0) = -\sum_{z_I} \operatorname{Res}_{z=z_I} \frac{\hat{A}_n(z)}{z} + B_n$$

where $z_I = -\dfrac{P_I^2}{2 P_I \cdot R_I}$ for I nontrivial (at least 2, no more than $n-2$ s.t) subset of momenta $\{p_i\}$
$P_I^2 = (\sum \hat{p}_i)^2$, $R_I = \sum r_i$

in the case $B_n = 0$ ($\lim_{z\to\infty} \hat{A}(z) = 0$)

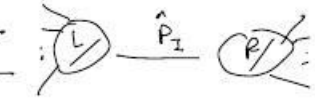
$$\hat{P}_I^2 = -\frac{P_I^2}{z_I}(z - z_I). \leftarrow \text{poles}$$

clearly, at z_I the propagator $\dfrac{1}{\hat{P}_I^2}$ goes on-shell. In that limit the shifted amplitude factorizes

$$\operatorname{Res} \hat{A}_n(z) = \hat{A}_L(z) \frac{1}{\hat{P}_I^2} \hat{A}_R(z_I) = \text{(L)} \frac{\hat{P}_I}{} \text{(R)}$$

$z = z_I$ ——— $\frac{1}{P_I^2}$

Then, one can write

$$A_n = \sum_{\substack{\text{diagrams} \\ I}} \hat{A}_L(z_I) \frac{1}{P_I^2} \hat{A}_R(z_I) = \sum_I$$ 

Sum over all possible factorization channels!

BCFW recussion is a conseque of just 2 shifted momenta!

That means any physical scattering amplitude must have
- convenient mass dimensions with coupling constant
- be lorentz invariant, ex: 4-point amp of scalrs must be a function of s, t, u
- it should have kinametic singularities which evoke locality ex: in 4-point amplitudes we have poles like $1/s$, not $1/s^2$.

So, A_n is a function of the kinematic variables with the above properties which one in general can write as

$$A_n = \int_{X_{\geq 0}} (\text{regulator}) \; \Omega(X_{\geq 0})$$

↑ for some form

Using some change of coordinates (we will talk about later for special theorem) one get a geometric shape defined by inequalities from the external kin and due to the BCFW relation we get poles at $P_I^2 = 0$ (kinematical) which should be equivalent to a restriction on one of the faces! the restriction has again to be an amplitude in the above sense!

Now, if we have two amplitudes that factorize in the same way. They must exactly the same. In that sense, two forms that restricts to the same values on t Kinametic boundries must be the same make this form unique.

For a reigon $Y \subset \mathbb{R}^N$ there is no guarantee that a form with the correc poles would be unique. However, that would be the case if we are workin in $\mathbb{P}^N$ since there are no - non trivial global holomorphic function. $\mathbb{P}^N$.

Thus, we are motivated to define the following:

Def: let X be an irreducible complex projective variety
- $X_{\geq 0}$ a d-dim closed semi-algebric subset of real points $X(\mathbb{R})$

$\subseteq X$ s.t $X_{>0} = \text{int}(X_{\geq 0})$ (w.r.t analytic topology on $X(\mathbb{R})$)

- $\text{int}(X_{>0})$ is an oriented real manifold.
- zariski closure $\partial X_{\geq 0}$ is a divisor on X denoted by $Y = Y_1 \cup \dots \cup Y_r$

Then, let $(Y_i)_{>0} = \text{int}(X_{\geq 0} \cap Y_i)$ on $(Y_i)_{\geq 0}(\mathbb{R})$

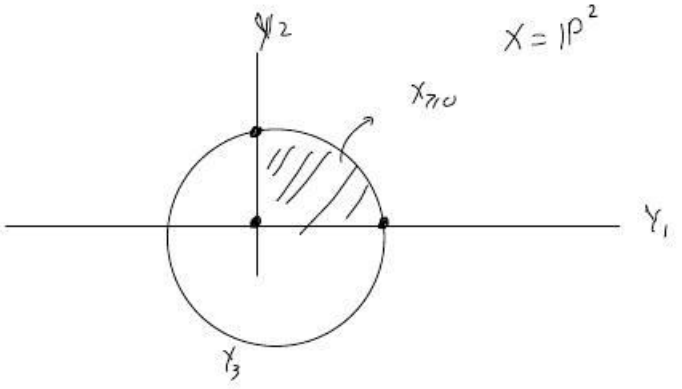
This pair $(X, X_{>0})$ is called a positive geometry if it's equipped with d-form on smooth manifold $X \setminus \text{Sing}(X)$ satisfying

- If $d=0$ $X = X_{\geq 0} = pt$ $\omega(X_{\geq 0}) = \pm 1$ (orientation of X)
- If $d > 0$ $\omega(X_{\geq 0})$ has simple poles on $Y_1, \dots, Y_r$ $Y_i \notin \text{Sing}(X)$ and no other poles on $X \setminus \text{Sing}(X)$

moreover, $(Y_i, (Y_i)_{\geq 0})$ are $(d-1)$ positive geometries with canonical form

$$\omega((Y_i)_{\geq 0}) = \text{Res}_{Y_i}\, \omega(X_{\geq 0})$$

Visual example



Rmk: for ω a d-form on X with simple pole along prime divisor $Y_i \subset X$ not c in $\text{Sing}(X)$, locally at a smooth point of $Y_i \setminus \text{Sing}(X)$

$$\omega = \eta \wedge \frac{df}{f} + \eta'$$

η η' are with no poles on Y_i and f local coords vanishes along Y_i. $\text{Res}_{Y_i}\omega = \eta|_{Y_i}$

$$\omega = \frac{g}{f_1 \cdots f_r}\, dx_1 \wedge \cdots \wedge dx_d = \frac{g}{f_1 \cdots \hat{f}_i \cdots f_r \frac{\partial f_i}{\partial x_d}}\, dx_1 \wedge \cdots \wedge dx_{d-1} \wedge \frac{df_i}{f_i}$$

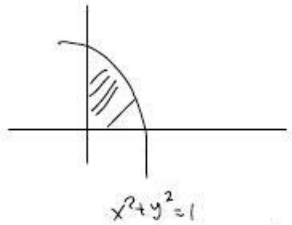
(Ex 1) $X = \mathbb{P}^1$, $X_{\geq} = [0, \infty]$

$\omega = \frac{dx}{x}$ ← has poles at $\{0, \infty\}$

$$\text{Res}\,\omega|_0 = 1 \qquad \text{Res}\,\omega|_\infty = -\frac{d(1/u)}{(1/u)}\Big|_\infty = -1$$

for $[a,b]$ we get $\frac{b-a}{(x-a)(b-x)}\,dx$

(Ex 2)



$$\omega = \frac{(1+x+y)}{xy(1-x^2-y^2)}\, dx \wedge dy$$

now we just need if this canonical form restricts to that of a line

- $\operatorname{Res} \omega|_{X=0} = \operatorname{Res}\left(-\frac{1+x+y}{y(1-x^2-y^2)}\, dy \wedge \frac{dx}{x}\right) = -\frac{1+y}{y(1-y^2)} = -\frac{dy}{y(1-y)}$

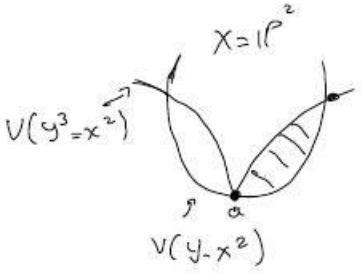
- $\operatorname{Res} \omega|_{Y=0} = \operatorname{Res}\left(\frac{1+x+y}{x(1-x^2-y^2)}\, dx \wedge \frac{dy}{y}\right) = \frac{1+x}{x(1-x^2)}\, dx = \frac{dx}{x(1-x)}$

- $\operatorname{Res} \omega|_{Y_3} = \operatorname{Res}\left(\frac{1+x+y}{xy(-2y)}\, dx \wedge \frac{d(1-x^2-y^2)}{1-x^2-y^2}\right) = -\frac{1+x+y}{2xy^2}$

$$= \frac{dx}{y(1-x-y)}\Big|_{Y_3}$$

Ex 3)

$$\omega = \frac{x^2+y^2+xy+x}{(y-x^2)(x^2-y^3)}\, dx \wedge dy$$



$$\operatorname{Res} \omega|_{V(y^3-x^2)} = \frac{x^2+y^2+xy+x}{(-3y^2)(y-x^2)}\, dx\Big|_{V(y^3-x^2)}$$

↑ This must be the residue of a line segment

but $a \in \operatorname{Sing}(V(y^3-x^2))$ so, it doesn't strictly satisfy the definition

To resolve this cusp singularity one just uses a normalization

$$\pi : [0,1] \longrightarrow Y_2$$
$$t \longmapsto (t^3, t^2)$$

one then just requires that the pullback $\pi^*(\eta|_Y)$ is the canonical form on $[0,1]$

$$\pi^*(\eta|_Y) = \frac{1}{t(t-1)}\, dt$$ ✓

Ex 4)



$X=\mathbb{P}^2$ nodal cubic

$V(y^2 - x^2(x+1))$, $x \leq 0$ $\quad \omega = \frac{2}{y^2 - x^2(x+1)}\, dx \wedge dy$

≤ 0

$$\operatorname{Res} \omega|_Y = \frac{dx}{y}$$

The problem here is that $(Y_{\geq 0})$ has no boundary!

to fix this we blow up at $(0,0)$

$$\pi : X' \longrightarrow X$$

s.t $X'_{\geq 0} = \overline{\pi^{-1}(\operatorname{int}(X_{\geq 0}))}$

$X'_{\geq 0}$ has two thing the exptual divisor Y_0, strict transform Y' of Y

put $y = zx \Rightarrow x^2(z^2 - x - 1) = 0$

$Y_0 \Leftrightarrow x = 0 \Rightarrow y = 0$
$Y' = (z^2 - 1, z^3 - z, z)$

So, we require that $(X', X'_{\geq 0})$ after blow up is a positive geometry

$$\omega = \frac{z}{z^2 - x - 1} \cdot \frac{dx}{x} \wedge dz$$

$\text{Res}\ \omega|_{Y_0} = -\frac{z}{z^2-1}\, dz$ ✓ canonical form on the line segment

Res w/ y we use the real parametrization $t \mapsto (t^2 - 1, t^3 - t, t)$
w/ $[-1, 1]$

$$\left(\frac{dx}{y}\right)\Big|_{Y'} = \frac{z}{t^2-1}\, dt \quad ✓$$

Computing canonical form. in $\mathbb{P}^m$
[Arkani-Hamed, –]

1) from poles & zeros
2) triangulation
3) push forward maps
4) volume of dual geometry

① Recall, that the measure on $\mathbb{P}^m$ is the following
for a point in $\mathbb{P}^m$ $(x_0, \dots, x_m) = X$
$dX = (dx_0, \dots, dx_m)$
$d^m X$ wedge product of m copies of dX

Fact! $\langle X d^m X \rangle = \frac{1}{m!} \varepsilon_{I_0 I_1 \dots I_m} X^{I_0} dX^{I_1} \wedge \dots \wedge dX^{I_m}$

let us see as an example what happen if we go to an affine patch in $\mathbb{P}^2$

$$X = (x_0, x_1, x_2)$$

$$\langle X d^2 X \rangle = X_0 (dX_1 \wedge dX_2) - X_1 (dX_0 \wedge dX_2) + X_2 (dX_0 \wedge dX_1)$$

$X_0 \to 1 \Rightarrow dX_0 = 0 \quad dX_1 = dx \quad dX_2 = dy$

$$\Rightarrow \langle X d^2 X \rangle = dx \wedge dy$$

So, we want to construct a form Ω

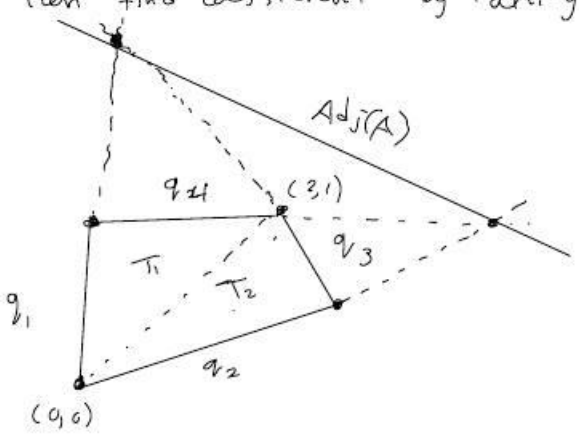
Ω has simple poles on Y_i so, Ω must look like

$$\frac{\text{Adj}(A)}{(\dots)}$$

$$\Omega(A) = \frac{Adj(A)}{\prod_{i=1}^{r} D(x)} \langle x\, d^m x \rangle$$

$$\Rightarrow \deg(Adj(A)) = \deg(D(x)) - m - 1$$

So, we write general form for $Adj(A)$ and then find coefficient by taking residues.



let A be a quadritral defined by

1. $x \geq 0$
2. $2y - x \geq 0$
3. $3 - x - y \geq 0$
4. $2 - y \geq 0$

$$\Omega = \frac{(A + Bx + Cy)\, dx\, dy}{x(2y - x)(3 - x - y)(2 - y)}$$

$$Res_{ij} = Res_j\, Res_i$$

$$\Rightarrow Res_{13} = \frac{A + 3C}{6} = 0$$

$$\Rightarrow Res_{24} = \frac{A + 4B + 2C}{12} = 0$$

$$\Rightarrow (A, B, C) = (12, -1, -4)$$

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(2) Def: we say $\{X_{i,\geq 0}\}$ triangulates $X_{\geq 0}$ if

1) $X_{i,\geq 0}$ is contained in $X_{\geq 0}$ and orientation agress
2) $X_{i,>0}$ are mutually disjoint (interior)
3) $\bigcup X_{i,\geq 0} = X_{\geq 0}$

Def: $\{X_{i,\geq 0}\}$ triangulates the empty set if $\forall x \in \bigcup X_{i,\geq 0}$ that doesnot lie on $\partial X_{i,\geq 0}$ we have

$$\#\{i \mid x \in X_{i,\geq 0} \text{ and } X_i \text{ positively oriented}\} = \#\{i \mid x \in X_i \text{ and } X_i \text{ is negatively or.}$$

Ex: $\{X_{1,\geq 0}, \ldots X_{r,\geq 0}\}$ interior triangles the empty set then,

$\{X_{2,\geq 0}, \ldots X_{r,\geq 0}\}$ interior triangles $X_{1,\geq 0}$

One can think of triangulations as summing over diagrams!

prop: If $\{X_{i,\geq 0}\}$ interior triangulates the empty set then

$$\sum \Omega(X_{i,\geq 0}) = 0$$

Thm: If $\{T_i\}$ triangles a polytope P then I look for a triangle

... polytope ...

$$\Omega(P) = \sum \Omega(T_i).$$

... triangle

$\Omega(\Delta^d) = \frac{dx_1}{x_1} \wedge \cdots \wedge \frac{dx_d}{x_d}$

$x_i = X_i / X_0$.

In previous example we have two triangles which give

$$\Omega(T_1) = \frac{c \, dx dy}{x(2-y)(y-2x)}$$

$$\text{Res } \Omega_1\big|_{x=0} = \frac{c \, dy}{y(2-y)} \Big\} \text{ of a line } c=2$$

$$\Omega(T_2) = \frac{c' \, dx dy}{(y-2x)(3-x-y)(2y-x)}$$

$$\text{Res } \Omega_2\big|_{2y-x} = \frac{c' \, dy}{y(1-y)(y)}$$

$$\Rightarrow c' = -9$$

- for orientation

then, one can see $\Omega = \Omega(T_1) + \Omega(T_2)$.

Convex polytopes in $\mathbb{R}^d$

Def: A convex polytope in $\mathbb{R}^d$ can be described by a list of affine linear inqualities in d-variables

$$P = \{ y \in \mathbb{R}^d : \underbrace{U \cdot y + z}_{\text{"} u_i \cdot y + z_i \text{"}} \geq 0 \}$$

$U \in \mathbb{R}^{n \times d}$ of rank d, $z \in \mathbb{R}^n$

An equivalent way of defining this is to do it projectively using $z_i = (1, z_i)$

$$P = \{ \sum_i c_i z_i \mid c_i > 0 \}$$

Furthermore, the polytope can be cut out of linear inqualities $Y \cdot w_j \geq 0$ where facets are given by $Y \cdot w_j = 0$.

Def: A dual polytope A^* in the dual projective space whose vertices are given by the dual vectors w_j.

$$A^* = \{ \sum_j c_j w_j \mid c_j > 0 \}$$

A^* is precisely the set cut out of the inqualities $w \cdot z_i \geq 0$

Thm: [ ] the canonical form of the polytope A is determined by the volume of the dual polytope A^*

$\Omega(A) = \ldots \; \mathrm{Vol}(A^*) \; \langle Y \, d^m Y \rangle$

$$JL(A) = Vol(A) \frac{\cdot \cdot \cdot \cdot}{m!}$$

one can further show that

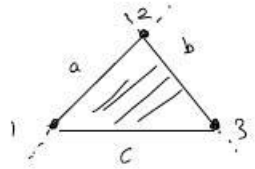
$$Vol(\Delta^*) = \frac{\langle w_1 \cdots w_{m+1} \rangle}{\prod^{m+1} (Y \cdot w_j)}$$

where $Y = c_1 w_1 + \cdots + c_m w_m + w_{m+1}$ $\overset{j=1}{!}$

if we return back,

$$\Omega(\Delta) = \frac{\langle z_1 \cdots z_{m+1} \rangle^m}{m! \prod \langle Y z_1 \cdots \hat{z}_i \cdots z_{m+1} \rangle} \langle Y \int^m Y \rangle$$

Ex:



$$= \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ 1 & 1 & 1 \end{vmatrix}$$

$$= \frac{1}{2} \frac{\langle z_1 \; z_2 \; z_3 \rangle}{(z_0 \cdot z_1)(z_0 \cdot z_2)(z_0 \cdot z_3)} \qquad z_0 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

where $z_{\cdot I} = \begin{pmatrix} x_i \\ y_i \\ 1 \end{pmatrix}$

using $z^2 w_I = 0$, $z_{iI} = \varepsilon_{IJK} w_c{}^J w_a{}^K$ ← intersection

similar for other

which give the formula $= \frac{1}{2} \frac{\langle abc \rangle^2}{\langle c, b, c \rangle \langle 0, a, b \rangle \langle 0, c, a \rangle} = [a\,b$

Apparently, this behavior of having simple poles on a divisor Y, looked similar to a mathematical structure of mixed hodge theory. we now give the definition of positive geometry as introduced by [Francis brow, dennt depart 2025]

Remark and results:

X is a smooth projective complex variety X, it's cohomology group enjoys the Hodge decomposition,

$$H^k(X, \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}$$

$H^{p,q}$ correspond to closed forms with exactly p holomorphic differentials, q anti-holomorphic differentials.

what if U which is a open variety (missing points or boundries)!
This does not work any more.
How did they fix this, they kinda embedded U inside a compact space and measured how badly it behaves near the missing boundary?

Def: Let X be a smooth complex manifold. A divisor $Y = \cup Y_i$ has simple normal crossings if each Y_i is a smooth subvariety and they intersect transversely. Locally near intersection point one can find complex coordinates $(z_1, \dots, z_n)$ s.t. Y is defined by

$$z_1 \cdot z_2 \cdots z_r = 0$$

(coordinate axes intersecting at the origin)

Thm [Hironaka, 1964] "resolution of singularities" Given $U \subset \mathbb{P}^n(\mathbb{C})$ be an alg hypersurface. Then, there exists a smooth compact variety X and a morphism $\pi : X \longrightarrow \mathbb{P}^n(\mathbb{C})$ such that

- π induces an isomorphism
$$X \setminus \pi^{-1}(V) \simeq \mathbb{P}^n(\mathbb{C}) \setminus U$$
- $\pi^{-1}(U)$ is a normal crossing divisor in X.

So, basically, given U an open variety, then $X \setminus Y \simeq U$

The idea "I think" is now to study what happens on X and track what happens on Y !! The compromise here is to look at forms on X that are allowed to have a specific type of singularity along Y.

Def: Let Y be a normal crossing divisor in X. A differential k-form ω on $X \setminus Y$ is said to have "logarithmic poles on Y if on local coordinates one can write ω as

$$\omega = \alpha \wedge \frac{dz_{i_1}}{z_1} \wedge \cdots \wedge \frac{dz_{i_n}}{z_n} + \beta$$

The sheaf is denoted by $\Omega_X^k(\log(Y))$.

To build a hodge str on the cohomology $H^k(V, \mathbb{C})$ pierre deligne introduced two filtrations

Def: The hodge filtration F, is a decreasing filtration of the spaces of forms with at least p holomorphic differential

$$F^p H^k(U, \mathbb{C}).$$

To track how bad things behave on the boundary he introduced the weight filtration $W_m H^k(U, \mathbb{C})$ is subspace of $H^k(U, \mathbb{C})$ generated by k-forms having at most $m-k$ logarithmic poles on Y.

- $W_k H^k$ k-forms with 0-poles (no blow up at the boundary)
- $W_{k+1} H^k$ smooth forms + forms with exactly 1 log pole at the boundary
- $\vdots$
- $W_{2k} H^k$ all forms up to k log poles. $\simeq H^k(U, \mathbb{C})$

$\cdots$ is k-forms with ... poles ... not ...

so, basically if is close forms with a certain pole count. we just quotient

$$\left.\begin{array}{l} gr_w H^k = \dfrac{W_w H^k}{W_{w-1} H^k} \end{array}\right\} \text{ have exactly } w-k \text{ poles.}$$

Thm: [Deligne] let U be a smooth complex alg variety choose a compactification X of U s.t $D = X \setminus U$ is a normal crossing divisor, then

- The space of forms on U with logarithmic poles along D does not depend on the choice of X
- All such forms are closed
- This induces an isomorphism

$$H^0(X, \Omega_X^k(\log Y)) \simeq \Omega^k_{\log}(U) \xrightarrow{\sim} F^k H^k(U, \mathbb{C})$$

Thm [poincaré duality] let X be a compact complex variety of dimension n and $Y \subset X$ a closed subvariety s.t $X \setminus Y$ is smooth.

since X is compact

$$H^\bullet(X, Y) \simeq H_c^\bullet(X \setminus Y)$$

the relative cohomology is isomorphic to the compactly supported cohomology the poincare duality pairing is

$$H_k(X, Y) \xrightarrow{\sim} H^{2n-k}(X \setminus Y)(n) \quad \text{(take twist)}$$

$$[\delta \overset{in}{\subset} Y] \longmapsto [\text{smooth closed } (n,n) \text{ form on } X \setminus Y \text{ (}\varphi\text{)}]$$

$$(2\pi i)^n \int_X \varphi \wedge \eta = \int_\delta \eta.$$

Thus, for X of dim k, we get

$$F^0 H_k(X, Y)_{\mathbb{C}} \simeq F^0 H^k(X \setminus Y)(k) = F^k H^k(X \setminus Y)_{\mathbb{C}}$$

Fun fact: The dual of a mixed hodge str H is the vector space $H^\vee = \mathrm{Hom}(H, \mathbb{Q})$ equipped with weigh filtration

$$W_{-n} H^\vee = (H / W_{n-1} H)^\vee = \ker(H^\vee \to (W_{n-1} H)^\vee)$$

and hodge filtration,

$$F^{-n} H^\vee_{\mathbb{C}} = (H_{\mathbb{C}} / F^{n+1} H_{\mathbb{C}})^\vee = \ker(H^\vee \to (F^{n+1} H_{\mathbb{C}})^\vee)$$

ex: duality between relative homology and cohomology | $h^{-p,-q}(H^\vee) = h^{p,q}(H)$

last but not least one considers the map

$$F^0 H_n(X,Y) \twoheadrightarrow \mathrm{gr}_0^W H_n(X,Y) = H_n(X,Y)/W_{-1}H_n(X,Y)$$

↑ cycles $\Leftrightarrow$ $(0,-q)$ forms

↑ is of weight zero! so only $(0,0)$

$$\mathrm{Ker} = \sum_{q=1}^{n} h^{0,-q} = \sum_{q=1}^{n} h^{q,0}$$

Def: The genus of (X,Y) is

$$g(X,Y) := \sum_{p>0} h^{-p,0}(H_n(X,Y))$$

Rmk: when X is smooth and compact $g(X) = \dim H^0(X, \Omega_X^n)$.

Def: let X be a compact complex variety of dimension n and $Y \subset X$ be a closed subvariety s.t $X \setminus Y$ is smooth. Assume that (X,Y) has genus 0. Then

$$\mathrm{can}: H_n(X,Y) \longrightarrow \Omega^n_{\log}(X \setminus Y)$$

be the composition

$$H_n(X,Y) \longrightarrow \mathrm{gr}_0^W H_n(X,Y) \xrightarrow{R^{-1}} \Omega^n_{\log}(X \setminus Y)$$

Def: [Brown, Dupont 2025] A positive geometry is a relative homology class $\sigma \in H_d(X,Y)$ of genus zero pair (X,Y).

To see if the form is canonical and unique we need to look at

Def: The combinatorical rank $\mathrm{cr}(X,Y) = \dim \mathrm{gr}_0^W H_n(X,Y)$

prop: let D be a normal crossing divisor in $\mathbb{P}^n(\mathbb{C})$. Then,

$$\mathrm{cr}(\mathbb{P}^n, D) = \binom{d-1}{n}$$

Rmk: this had rank 1 iff $d = n+1 \Rightarrow D$ give a simplex like positive geometry where the canonical form $(X, X_{>0})$ had no zero on X.

Rmk: two definitions are not equivalent

- For a pair $(X, X_{>0})$ $[X_{>0}] \in H_1(X,Y)$] the first definition is more

$x_{2,0}$ might not be contained in $X \subset (\mathbb{R})$
$w([x_{2c}])$ might be zero

restrictive!

one can also see that given [6] the process of taking massive residues completely determines the log form!

Now, we move on and try to informally describe the Amplituhedron!

spinor helicity formalism:

Kinematics are the heart of everything as we said

- Many scattering amplitudes are fixed by kinematic constraints like lorentz invariance and momentum conservation.

we have a characterization by $p_\mu = (p_0, \vec{p})$

The spinor helicity maps the components of the 4-vectors into 2x2 matrices

$$P_{\alpha\dot{\alpha}} = p_\mu \, \sigma^\mu_{\alpha\dot{\alpha}} \qquad \sigma^\mu = (1, \vec{\sigma})$$

$$\Rightarrow \quad \det(P) = p^\mu p_\mu = 0 \quad \leftarrow \text{(massless case)}$$

That means P has rank 1 $\Rightarrow$ $P_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}$
(holomorphic & anti-holopic spinors)

Def: $\langle ij \rangle = \lambda_{i\alpha} \lambda_{j\beta} \varepsilon^{\alpha\beta}$, $[ij] = \tilde{\lambda}_{i\dot{\alpha}} \tilde{\lambda}_{j\dot{\beta}} \varepsilon^{\dot{\alpha}\dot{\beta}}$
angle & square brackets.

claim: Any function of n-d kinematic data can be written exclusively in terms of these objects.

(Ex) $$s_{ij} = (p_i + p_j)^2 = 2 p_i p_j = \langle ij \rangle [ij]$$

since each spinor is 2-dimensional one can write any spinor as a linear combination of 2 independent spinors

$$\langle ij \rangle \lambda_k + \langle ki \rangle \lambda_j + \langle jk \rangle \lambda_i = 0$$

so, $A(p_i, \varepsilon_i^{h_i}) = A(\lambda_i, \tilde{\lambda}_i, h_i)$

$\not\Rightarrow$ due to momentum conservation $\sum_i \lambda_i \tilde{\lambda}_i = 0$

with identification $p_i = \lambda_i \tilde{\lambda}_i$, $\varepsilon_i^+ = \sqrt{2} \, \frac{r_i \tilde{\lambda}_i}{\langle r i \rangle}$, $\varepsilon_i^- = -\sqrt{2} \, \frac{\lambda_i \tilde{r}_i}{[r i]}$

$n_i, \tilde{n}_i$ are arbitrary vectors

let us consider for example 3-point amplitude



little group $\longrightarrow$ $t_1^{-2h_1} t_2^{-2h_2} t_3^{-2h_3} A$

$$A(1^{h_1} 2^{h_2} 3^{h_3}) = \langle 12\rangle^{n_3} \langle 23\rangle^{n_1} \langle 31\rangle^{n_2}$$

by matching factors,

$$\left.\begin{aligned} n_3 + n_2 &= -2h_1 \\ n_3 + n_1 &= -2h_2 \\ n_1 + n_2 &= -2h_3 \end{aligned}\right\} \Rightarrow \begin{aligned} n_1 &= h_1 - h_2 - h_3 \\ n_2 &= h_2 - h_3 - h_1 \\ n_3 &= h_3 - h_1 - h_2 \end{aligned}$$

from $\sum \tilde{\lambda}_i^{\dot\alpha} \lambda_i^{\alpha} = 0$

$|1\rangle[1| + |2\rangle[2| + \cdots = 0$

$\langle 1|$

$\Rightarrow \langle 12\rangle[2| + \cdots = 0$

if $\tilde\lambda$ are independent

$\Rightarrow \langle ij\rangle = 0 \quad \forall\, i,j$

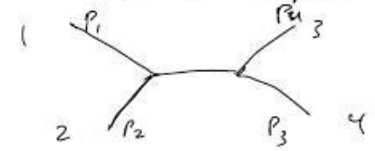
if λs are independent

$\Rightarrow [ij] = 0$

$$A(1^{h_1} 2^{h_2} 3^{h_3}) = \langle 12\rangle^{h_3 - h_1 - h_2} \langle 23\rangle^{h_1 - h_2 - h_3} \langle 31\rangle^{h_2 - h_3 - h_1}$$
$$= [12]^{h_1 + h_2 - h_3} [23]^{h_3 + h_2 - h_1} [31]^{h_1 + h_3 - h_2}$$

To do dynamics we can employ factorization properties

let us take 4-particle amplitude



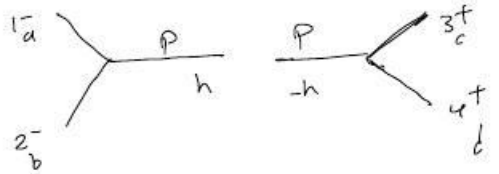
$$\lim_{s\to 0} s A_4 = A_3 A_3$$

$\Rightarrow [A_4] = 2[A_2] - 2$ of mass dimension

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if we consider

$$A_3(1^-, 2^-, 3^+) = C \underbrace{\frac{\langle 12\rangle^3}{\langle 23\rangle\langle 31\rangle}}_{\text{mass dimension } 1} \Rightarrow [A_4] = 0$$



$$A_4 \longrightarrow t_1^2 t_2^2 t_3^{-2} t_4^{-2} A_4$$

$$A_4 = \langle 12\rangle^2 [34]^2 F(s,t,u)$$

$\langle 12\rangle, [34]$ have dim 1 $\Rightarrow$ $F(s,t,u)$ has dim 2

must be log in s,t,u — which are contractions of p_s

$$\Rightarrow A_4 = \langle 12\rangle^2 [34]^2 \left(\frac{c_{st}}{st} + \frac{c_{su}}{tu} + \frac{c_{tu}}{su} \right)$$

$(t = -u)$ since $s=0$ $\quad s+t+u=0$

$$\operatorname*{Res}_{s\to 0} (sA_4) = \langle 12\rangle^2 [34]^2 \frac{1}{t} (c_{st} - c_{su})$$

$$= \sum_e f_{abe} f_{cde} \frac{\langle 12\rangle^3}{\langle 1P\rangle\langle P2\rangle} \cdot \frac{[34]^3}{[3P][P4]} = \sum_e f_{abc} \langle 12\rangle^2 [$$

we know, $P = -(P_1 + P_2) = P_3 + P_4$

$= -|1\rangle[1| - |2\rangle[2|$

$= |3\rangle[3| + |4\rangle[4|$

$\langle 1P\rangle[P4] = \langle 1|P|4]$

$= -\langle 12\rangle[24]$

$= \langle 13\rangle[34]$

simply $\langle P2\rangle[3P] = \langle 42\rangle[54]$

repeating this one can find

$$\left.\begin{aligned} C_{st} - C_{us} &= \sum f_{abc} f_{cde} \\ C_{tu} - C_{st} &= \sum f_{bce} f_{ade} \\ C_{us} - C_{tu} &= \sum f_{cae} f_{bde} \end{aligned}\right\} \text{ structure constants}$$

Thm: using an induction argument for unordered particles one can find that

$$A_n(1^+ 2^+ \dots i^- \dots j^- \dots n^+) = \frac{\langle ij\rangle^4}{\langle 12\rangle\langle 23\rangle \cdots \langle n1\rangle} \quad \text{(MHV)}$$

(Parke-Taylor)

Rmk: it turns out that

- for $n>3$, helicity amplitudes with $h = \sum_i h_i$ satisfying $|h| \geq n-2$ vanish. The simplest non-vanishing amplitudes are those with $h = n-4$. (MHV)
- $\exists$ $\overline{\text{MHV}}$ anti-maximally helicity violating $h = 4-n$.

There are also other non-vanishing amplitude called N^kMHV amplitudes with total helicity of $h = n-4-2k$.

we focus on $n=6$, $k=1$ $\Rightarrow h=0$

There are 3 types of these (non-vanishing) amplitudes

Define: $S_{ijk} = [ij]\langle ij\rangle + [ik]\langle ik\rangle + [jk]\langle jk\rangle$

$[i|P_{jkl}|L\rangle = [ij]\langle jL\rangle + [ik]\langle kL\rangle$

- $A(1^- 2^- 3^- 4^+ 5^+ 6^+) = \frac{[4|P_{23}|1\rangle^3}{[23][34][2|P_{34}|5\rangle} \frac{1}{\langle 56\rangle\langle 61\rangle S_{234}}$

$$+ \frac{[6|P_{54}|3\rangle^3}{[61][12][2|P_{54}|5\rangle\langle 34\rangle\langle 43\rangle} \frac{1}{S_{612}}$$

• $A(1^+, 2^-, 3^+, 4^-, 5^+, 6^-) = \dfrac{[15]^4 \langle 24\rangle^4}{[56][61]\langle 1|P_{23}|4\rangle [5|P_{43}|2\rangle \langle 23\rangle\langle 54\rangle} \dfrac{1}{s_{234}}$

$+ (i \to i+2 \bmod 6) + (i \to i+4 \bmod 6)$

• $A(1^+, 2^+, 3^-, 4^+, 5^-, 6^-) = \dfrac{[4|P_{12}|3\rangle^4}{[45][56][4|P_{32}|1\rangle\langle 6|P_{12}|5\rangle\langle 12\rangle\langle 23\rangle s_{123}}$

$+ \dfrac{[24]^4\langle 56\rangle^3}{[23][34][2|P_{34}|5\rangle [4|P_{52}|1\rangle\langle 61\rangle s_{234}} + \dfrac{[12]^3\langle 35\rangle^4}{[61][2|P_{34}|5\rangle [6|P_{54}|3\rangle\langle 34\rangle\langle 45\rangle s_3}$

Spinor helicity $\longrightarrow$ Momentum twistors

« little group scaling represents exactly the same momentum » $\lambda \to t\lambda$

for $n \geq 4$ with helicity variables $\{\lambda_i, \tilde{\lambda}_i\}$ satisfying

$\langle i\ i+1\rangle \neq 0$, $[i\ i+1] \neq 0 \quad \forall i$

Def: A twistor is a point in $\mathbb{CP}^3$ which we denote by

$$Z = (z^1 : z^2 : z^3 : z^4)$$

Thm: let $\{\lambda_i, \tilde{\lambda}_i\}$ be a configuration of $n \geq 4$ particles. Then, there exist n twistors $Z_i = (\lambda_i^1 ; \lambda_i^2 : \mu_i^1 : \mu_i^2) \in \mathbb{CP}^3$ and n dual twistors $\tilde{Z}_i = (\tilde{\mu}_i^1 ; \tilde{\mu}_i^2 : \tilde{\lambda}_i^1 : \tilde{\lambda}_i^2) \in (\mathbb{CP}^3)^*$ satisfying,

1. $\forall i$, the plane defined by $\tilde{Z}_i$ contains Z_{i-1}, Z_i, Z_{i+1}
2. $\forall i$, Z_i, Z_{i+1}, Z_{i+2} and Z_{i+3} are not contained in a plane

[Penrose]

where $\mu^{\dot\alpha} = x^{\dot\alpha\alpha}\lambda_\alpha$ ← penrose incidence relation

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Rmk: So a spinor can be extended to a twistor by adding the μs in particular

$$\langle ij\rangle = Z_i^T I Z_j \qquad [ij] = \tilde{Z}_i^T \tilde{I} \tilde{Z}_j$$

$$I = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \qquad \tilde{I} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

? infinity twistor

• one can find $\tilde{Z}_j^A = \dfrac{\varepsilon^{ABCD} Z_{i-1}^B Z_i^C Z_{i+1}^D}{\langle i-1\ i\rangle\langle i\ i+1\rangle}$ ↷ (plane $\tilde{Z}_i$ containing Z_{i-1}, Z_i, Z_{i+1})

this is ad twistors coordinates

Def: $\langle i\,j\,k\,l\rangle = \det\left(z_i\, z_j\, z_k\, z_l\right) = \varepsilon_{ABCD}\, z_i^A\, z_j^B\, z_k^C\, z_l^D$

Thm: [Hodges, 2013] let $\{z_i\}$ be a collection of $n \geq 4$ points in $\mathbb{CP}^3$ th satisfies $\langle i\, i+1\rangle \neq 0$, $\langle i\, i+1\, i+2\, i+3\rangle \neq 0$ $\forall i$ then for $a = 1, 2$

$$(*) \qquad \lambda_i^a = z_i^a \quad , \quad \tilde{\lambda}_i^a = \frac{\langle i-1\, i\rangle\, z_{i+1}^{a+2} + \langle i+1\, i-1\rangle\, z_i^{a+2} + \langle i\, i+1\rangle\, z_{i-1}^{a+2}}{\langle i-1\, i\rangle\langle i\, i+1\rangle}$$

(Ex) one can use these relations to show for example

$$[i\ i+1] = \frac{\langle i-1\ i\ j+1\ i+2\rangle}{\langle i-1\ i\rangle\langle i\ i+1\rangle\langle j+1\ i+2\rangle} .$$

one can the find that ;

$$[4|P_{56}|1] = \frac{\langle 1345\rangle}{\langle 34\rangle\langle 45\rangle} \quad , \quad [6|P_{12}|3\rangle = \frac{\langle 1356\rangle}{\langle 56\rangle\langle 61\rangle}$$

$$[2|P_{34}|5\rangle = \frac{\langle 1235\rangle}{\langle 12\rangle\langle 23\rangle} \qquad S_{612} = \frac{\langle 2356\rangle}{\langle 23\rangle\langle 56\rangle} \qquad S_{234} = \frac{\langle 1245\rangle}{\langle 12\rangle\langle 45\rangle}$$

$\Rightarrow$

$$A(1^-, 2^-, 3^-, 4^+, 5^+, 6^+) = \frac{\langle 12\rangle^4\langle 23\rangle^4}{\langle 12\rangle\langle 23\rangle\langle 34\rangle\langle 45\rangle\langle 56\rangle\langle 61\rangle} \times \frac{1}{\langle 1235\rangle}$$

$$\times \left(\frac{\langle 1356\rangle^3}{\langle 2356\rangle\langle 1236\rangle\langle 1256\rangle} + \frac{\langle 1845\rangle^3}{\langle 1234\rangle\langle 1245\rangle\langle 2345\rangle} \right)$$

& all the others !

Things seem to get complicated one by one . To collect all these amplitud physisfs use supersymetric amplitudes of $N=4$ Yang mills theory . we can now think of it as a book keeping formula to assemblee all

$\binom{n}{k+2}$ n-particle amplitudes into a singel thing .

$N^k MHV$

Def: A Grassmann variable (anti commuting) θ_a if $\theta_a\theta_b = -\theta_b\theta_a$ S.t integration over θ's is defined in the sence of Berzin by

$$\int d\theta_- = 0 \qquad \int \theta_- d\theta_- = 1$$

Def: A momentum supertwistor is a point in $\mathbb{CP}^{3}$ $(z^1 : z^2 : z^3 : z^4 : x^1 : x^2 : x^3 : x^4)$

Z^A are commuting coordinates, x^A are grassman variables

There is a straightforward extension of (*) to $\mathbb{CP}^3$. with $\langle i\ i+1\rangle \neq 0$ and $\langle i\ i+1\ i+2\ i+3\rangle \neq 0 \quad \forall i$, the pieces (2) formulas are satisfied with

$$\eta_i^A = \frac{\langle i-1\ i\rangle x_{i+1}^A + \langle i+1\ i-1\rangle x_i^A + \langle i\ i+1\rangle x_{i-1}^A}{\langle i-1\ i\rangle \langle i\ i+1\rangle}$$

where η_i satisfy something called the supermomentum conservation

$$\sum_i \lambda_i^\alpha \eta_i^A = 0$$

notation: $\bar{\eta}_i = \prod_A \eta_i^A$, $d\bar{\eta}_i = \prod_A d\eta_i^A$. $d\bar{\eta} = \prod_i d\bar{\eta}_i$

MHV superamplitudes

claim: All MHV amplitudes are contained in the MHV superamplitude

$$A_n^{MHV} = \frac{\delta^{(8)}(q)}{\langle 12\rangle\langle 23\rangle \cdots \langle n1\rangle}$$

$$\delta^{(8)}(q) = \prod_{A=1}^{4} \sum_{i<j} \langle ij\rangle \eta_i^A \eta_j^A$$

with, $A_6(\cdots i^- j^-) = A_n \big|_{\bar{\eta}_i \bar{\eta}_j}$

$$= \int \left(\prod_{l \neq i,j} \bar{\eta}_l\right) A_n^{MHV}\, d\bar{\eta} \quad \leftarrow \text{Berezin integral}$$

Apparently for some weird algebra, the NMHV superamplitudes

$$A_n^{NMHV} = A_n^{MHV} \sum_{j=2}^{n-2} \sum_{k=i+2}^{n-1} R_{nj|k} \quad \leftarrow \text{dual superconformal invariants}$$

for $n=6$, this is the important thing

$$A_6^{NMHV} = A_6^{MHV} \left([61234] + [61245] + [62345] \right)$$

where,

$$[abcde] = \frac{\prod_{A=1}^{4} \left(\langle abcd\rangle x e^A + \text{cyclic perm} \right)}{}$$

$\langle abcd\rangle\langle bcde\rangle\langle cdea\rangle\langle deab\rangle\langle eabc\rangle$

we have 20 expressions here $\binom{6}{3} = 20$ which you can find again by doing the similar Berzin integral

$$A^{(6)}_{i,j,k} = \int \prod_{L+i,j,k} d\bar{\eta}_L \; \mathcal{A}_6^{\mathcal{N}\text{MHV}} \, d\bar{\eta} \qquad \in \begin{pmatrix}\text{this is very}\\ \text{complex}\end{pmatrix}$$

Before introducing the amplidron. [Hodges, 2013] realized that the above expression can is a volume of simplices in the ~~$\mathbb{R}$~~ projective spaces. Then, Arkani-Hamed 2013 shows it's close connection to the amplihedron 2013.

Recall that:
let V be a vector space with basis $\{e_i\}$ $\overset{\text{for}}{\wedge}$ any k linearly independent vectors $\{v_1, \dots, v_k\}$ we can associate a linear subspace $\langle v_1, \dots v_k\rangle$ in V.

Rmk: this process is not injective since any linear combination of V is exactly the same which is the left multiple by a matrix in $GL(k)$ of
$M = (v_1, \dots, v_k)^T$

pullback embeddings

$$i: \left\{ \begin{matrix} k \text{ linear} \\ \text{subspaces of} \\ V \end{matrix} \right\} \longrightarrow \mathbb{P}\left(\wedge^k \binom{n}{k}\right)$$

$$\mathcal{M}_{k\times N} / GL(k)$$

just like we can identify a plane in $\mathbb{R}^3$ by a vector up to scalling ($v \propto \det(v_1, v_2, \cdot)$) we can identify higher subspaces by forming a vector from $k\times k$ minors.

Def: the image of i is known as the Grassmanian $G(k,n)$

Thm: $G(k,n)$ is zariski closed & irreducible of dim $k(n-k)$

Def: The totally positive grassmann $G^+_{\geq 0}(k,n) \subset Gr(k,n)$

whose points can be represented by $k\times n$ matrix with strictly positive maximal minors $\Delta_I(A) > 0$

let $Z: \mathbb{R}^n \longrightarrow \mathbb{R}^{k+m}$ s.t $n \geq k+m$ be a linear map
let V be a k-dim subspace $V \subset \mathbb{R}^n$ then,
$Z(V) \subset \mathbb{R}^{k+m}$ is at most k-dim

This induces a rational map
$Z: Gr(k,n) \longrightarrow Gr(k, k+m)$

$$[x] \longmapsto [x.Z]$$

where Z is a real matrix $\in \mathbb{R}^{n \times (k+m)}$

Def: Assuming Z is well-defined on $Gr(k,n)_{\geq 0}$ the image

$$A_{n,k,m} = Z(Gr(k,n)_{\geq 0}) \quad \text{is called the Amplituhedron}$$

In our case $n=6$, $k=1$, $m=4$

$Z \in \mathbb{R}^{6\times 5}$ matrix and we are living in $\mathbb{P}^4$

The recipe: (momentum twistor; grassmannian)

- define $Z_i = (Z_i^1, Z_i^2, Z_i^3, Z_i^4, \phi^A, \chi_i^A)$
- $A_{6,1,4}(Z)$ turns out to be a cyclic polytope which can be triangulated into 3 simplices

$$\Delta_{\{61234\}}, \Delta_{\{61245\}}, \Delta_{\{62345\}}$$

$$\Delta_{\{abcde\}} = \text{Convex hull}\{Z_a, Z_b, Z_c, Z_d, Z_e\}$$

- we have a symbol we know how to get it's canonical form for $Y = (0:0:0:0:1)$

$$\Omega_{\{abcde\}}(Y;Z) = \frac{\langle abcde\rangle^4}{\langle Yabcd\rangle\langle Ybcde\rangle\langle Ycdea\rangle\langle Ydeab\rangle\langle Yeab\rangle}$$

- $\int \Omega^{(Y_0,Z)}_{\{abcde\}} d^4\phi$ & sum $= [abcde]$

↑ volume of the simplex which exactly matches that in $N=4$ SYM